

Benchmark Implementation Guidelines

This supplementary section provides implementation guidance for colony cognitive security benchmarks introduced in Part 1, Section S05.

Test Environment Specification

Colony CogSec benchmarks require test environments that support:

- 1. **Scalable agent populations** — $n \in \{10, 50, 100, 500, 1000\}$
- 2. **Configurable stigmergic substrates** — Shared memory, message queues, artifact stores
- 3. **Instrumented communication channels** — Full message logging with timestamps
- 4. **Controllable adversary injection** — Precise Sybil insertion and signal poisoning
- 5. **Collective function measurement** — Aggregate outcome metrics beyond individual agent states

Table 1: Recommended colony CogSec benchmark configurations.

Benchmark	Min n	Stigmergy	Adversary	Duration
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